

Evidence for Success of Microbat Rehabilitation in Australia:

Summary and Recommendations for Bat Rehabilitators

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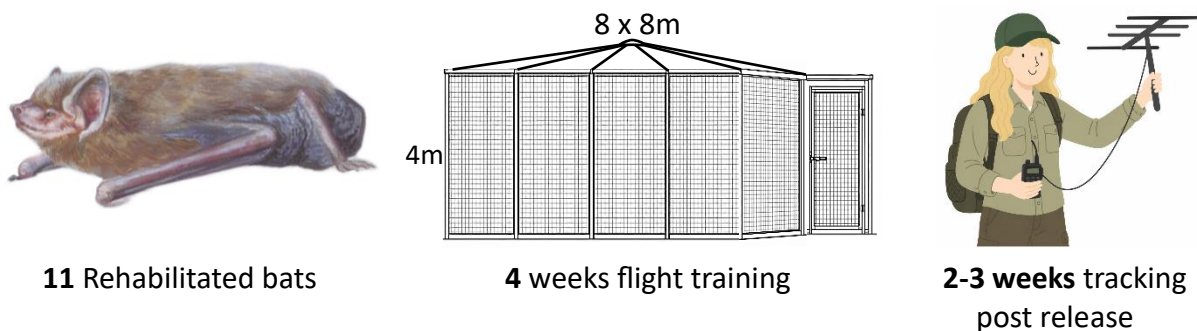
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For the first time in the Southern Hemisphere, researchers have confirmed that rescued and rehabilitated microbats, (including hand-raised orphans) can successfully return to the wild by radiotracking them after release. Bats were found to successfully forage, find roosts, and re-join their colonies. 8/11 (72%) of the released bats were located after release back in their home roost or a roost nearby and tracked for between 2-18 days. Not all bats were able to be located after release, but the majority of bats were tracked long enough to show they can catch food in the wild and find roosts, which is great news for bat carers! Basically – the current rehabilitation protocols worked!

WHAT WE DID

Eleven rehabilitated bats were flight-trained for four weeks, released at their rescue sites, and radio-tracked for up to three weeks to determine whether they survived and reintegrated with wild colonies.



Eleven broad-nosed bats (*Scotorepens spp*) from South East Queensland were rehabilitated using the protocols described in the *Australian Microbat Manual*, and radio-tracked after release in Autumn 2025. The bats came from a range of backgrounds: six were hand-raised from birth, two were juveniles born to injured mothers in care, and three were adults with injuries. All went through a standard rehabilitation program before spending four weeks flying freely in a large outdoor flight cage. Bats were released back to their site of rescue and radio-tracked with tags monitoring skin temperature to track survival until the tag fell off or the battery of the tag was exhausted (around 2/3 weeks). Whenever we identified a roost, we watched the entrance at dusk with to identify if the bats reintegrated with a colony, and trapped at two roosts we were able to identify and safely access to check if the rehabilitated bats were healthy.

WHAT WE FOUND

Eight of the eleven bats were tracked for over 4 days (which means they were able to survive past what we expect if they couldn't catch wild insects). Bats used an average of two roosts during the time they were tracked, but one ambitious bat moved roosts 5 times during the trial. Bats switched between trees and building roosts, all less than one kilometre from their original roost.

Two were recaptured 10/11 days after release at their roost, and were in good body condition. Both had removed their tags almost immediately, but we were able to confirm they had reintegrated with a colony, and had maintained good condition with no signs of injury or disease.

No deaths or adverse events were recorded during the trial. This means that none of the bats we tracked were found grounded or injured during the trial.

Three bats weren't able to be located after release. While we can't say for sure what happened, there are a few possibilities. Based on our finding that the two recaptured bats had removed their tags, we suspect what happened is the glue on the tags failed early, and the tag was damaged by the bat as it removed them. Bats could have also flown out of detection range, and it is possible the bats died, but we should have been able to pick up a signal from the tag even if the bat had passed away. Radiotracking can be difficult in urban areas due to interference from buildings, and studies in the same area have about a 50% success rate relocating bats with tags, so being able to locate 8/11 bats is actually an above average success rate!

Bats that we were able to track foraged nightly, switched roosts between buildings and trees, and reintegrated with colonies of their own species — even hand-raised orphans that had never encountered natural roosts or wild insects before.

Compared to the success rate of previous equivalent rehabilitation trials in Europe (Kelly et al., 2008, 2012; Serangeli et al., 2012), **we had similar success rates, and less adverse events.** We suspect the success in this trial was because bats had access to a large flight aviary, which is likely crucial for getting sufficient training. Releasing bats to their site of rescue may also have allowed them to quickly reunite with their home colony.

Like all research, this study had some limitations. First, it relied on radio trackers which have very limited battery life, and can fail if bats remove the tag or fly out of detection range. Second, we only focused on broad-nosed bats, so we don't know how successful other species may be after rehabilitation, although we have no reason to assume results wouldn't apply to other species.

RECOMMENDATIONS

1. Release bats at their exact rescue site if you can, even if the mother has died or you're not sure if the roost is still there.

Tracked rehabilitated bats selected multiple roosts within 1 km of their rescue site, which is a pretty tight home range. Even if the original roost is out of use or destroyed, it should be assumed that the colony will have other roosts nearby, as almost all microbats in Australia use a network of roosts within their home range. Returning bats to their home range will likely allow them to reintegrate with their relatives, which is thought to be important for their survival. All of the orphan bats we were able to locate with tracking returned to their original roost and re-joined a colony. In cases where no rescue site is known, there is some good news- comparable studies in Europe which released bats to new locations had some failed releases, but survival rates weren't significantly different without our trial. So, it's likely that releasing bats with no known address paired up with along with other members of their species is still a viable backup strategy.

2. Prioritise extended time in a large flight cage before release

Four weeks of free-flying in a large outdoor enclosure (this study used an 8×8×4 m aviary) appears to be the key factor in release success. Previous studies found that bats given only small rooms or short flight times had lower recapture rates and multiple instances of failed releases. Microbat rehabilitators reported that many bats crash land or are reluctant to fly when they first come into the flight aviary, and it takes the full four weeks until they are confident fliers, so flight training is likely crucial for any bats needing longer rehabilitation (in care more than two weeks)

3. Bats raised on mealworm diets will still be able to forage on wild insects

All of the bats in this trial were fed on boosted mealworms coated in Missing Link or Wombaroo Small Carnivore Mix. Due to a very rainy rehabilitation period, very few wild insects were introduced into the outdoor aviary where they got their flight training – but we still found evidence that even hand-raised bats were able to successfully forage, as most (63%) of the bats we tracked were alive at least 4 days after release before the tags failed, longer than we would expect if they weren't able to hunt for food.

4. Long periods of time in rehab care are likely worth it.

Many of the bats in this trial were in care for long periods during recovery or while waiting for space in a flight aviary to open up. Multiple bats that spent over seven months in care were tracked long enough to confirm short term survival. Based on this evidence, as long as a bat can pass the four-week flight training (tested by confirming strong flights for at least three nights before release), we should continue trying to rehabilitate bats that need longer term care.

Table 1. Summary of all rehabilitated bat outcomes from release trial in South East Queensland.

ID	Species	Sex	Age	Months in care	Roost found	Roost mates	Days tracked	# Roosts	Outcome
950	<i>EBN</i>	F	A	6.8	Yes	Yes	10	1	Tag dropped inside roost, recaptured day 10
430	<i>EBN</i>	F	A	7.3	Yes	Yes	7	2	Signal lost
149	<i>EBN</i>	F	JC	5.4	Yes	Yes	6	5	Signal lost
550	<i>Sp.</i>	M	A	7.9	Yes	Yes	11	1	Tag dropped inside roost, recaptured day 11
390	<i>Sp.</i>	M	J	3.9	Yes	Yes	2	2	Tag presumed lost inside roost
028	<i>Sp.</i>	M	J	3.7	Yes	Yes	18	1	Battery depleted
110	<i>Sp.</i>	M	J	3.7	Yes	Yes	6	2	Signal lost
349	<i>Sp.</i>	F	J	3.9	Yes	Yes	11	2	Battery depleted
470	<i>Sp.</i>	F	J	4.9	No	NA	NA	NA	Signal lost
431	<i>Sp.</i>	F	J	4.4	No	NA	NA	NA	Signal lost
229	<i>EBN</i>	F	JC	5.4	No	NA	NA	NA	Signal lost

Species: EBN = Eastern broad-nosed bat (*Scotorepens orion*), Sp. = Central eastern broad-nosed bat (*Scotorepens sp.* [Paranby]). Age class: A = adult, J = juvenile, JC = juvenile born in care.

Sources and Relevant Readings

Kelly, A., Goodwin, S., Grogan, A., & Mathews, F. (2008). Post-release survival of hand-reared pipistrelle bats *Pipistrellus spp.* *Animal Welfare*, 17(4), 375–382.

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Lyons, R. (2020). *Introduction to the Care and Rehabilitation of Microbats (Version 6)*. Wildcare Australia Inc. (Current Rehab manual is on the [Australian Microbat Rehabilitation Forum Facebook Group](#), join to access the PDF)

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